

1. Administrative & Study Identification

Full Study Title: A Phase 2 Randomized Study of Relatlimab Plus Nivolumab in Combination With Chemotherapy vs. Nivolumab in Combination With Chemotherapy as First Line Treatment for Participants With Stage IV or Recurrent Non-small Cell Lung Cancer (NSCLC) **Short Title/Acronym:** RELATIVITY-104 **Protocol Number:** CA224-104 **NCT Number:** NCT04623775 **Posting ID:** 902426380423784548 **Sponsor Name:** Bristol Myers Squibb **Trial Status:** ACTIVE_NOT_RECRUITING **Investigational Product:** Relatlimab (anti-LAG-3 antibody) + Nivolumab (anti-PD-1 antibody) combined with Platinum Doublet Chemotherapy (PDCT: Carboplatin, Cisplatin [Platinol, ATC: L01XA01], Paclitaxel, Nab-Paclitaxel, or Pemetrexed) **Phase of Development:** Phase 2 **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety profile of relatlimab plus nivolumab in combination with platinum doublet chemotherapy (PDCT) in Part 1 and to determine if nivolumab plus relatlimab in combination with PDCT improves the Objective Response Rate (ORR) when compared to nivolumab plus PDCT in participants with previously untreated Stage IV or recurrent non-small cell lung cancer (NSCLC) in Part 2. **Secondary Objectives:** To evaluate the percentage of participants with TRAEs leading to discontinuation in Part 1, the number of participants with Treatment Related AEs in Part 1, and the number of participants with Treatment Related SAEs in Part 1, in addition to assessing Progression-Free Survival (PFS), Duration of Response (DOR), and ORR stratified by PD-L1 expression.

2.2 Background & Rationale To address the need for improved first-line treatment options in advanced NSCLC. The inhibition of PD-1 and Lymphocyte Activation Gene-3 (LAG-3) pathways with nivolumab plus relatlimab has shown strong synergy, restoring effector T-cell function and improving anti-tumor immune responses to a greater extent than either antibody alone. This study tests whether adding relatlimab to standard-of-care chemoimmunotherapy expands clinical benefits for patients with advanced NSCLC.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator / Placebo-controlled (nivolumab + placebo + PDCT) **Blinding:** Double-blind **Randomization:** Randomized 1:1

3.2 Planned Enrollment & Duration Target **\$N\$:** 468 enrolled participants **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 17-Feb-2021 **Estimated Primary Completion:** 11-Jan-2024 (Actual) **Estimated Study Completion:** 27-Feb-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically confirmed metastatic non-small cell lung cancer (NSCLC) of squamous (SQ) or non-squamous (NSQ) histology with Stage IV A/B (as defined by the 8th International Association for the Study of Lung Cancer Classification) or recurrent disease following multi-modal therapy for locally advanced disease. 2. Eastern Cooperative Oncology Group (ECOG) performance status (PS) of less than or equal to 1 at screening and confirmed prior to randomization. 3. Measurable disease by computed tomography (CT) or magnetic resonance resources (MRI) per response evaluation criteria in solid tumor version 1.1 (RECIST 1.1) criteria. 4. No prior systemic anti-cancer treatment (including epidermal growth factor receptor (EGFR) and anaplastic lymphoma kinase (ALK) inhibitors) given as primary therapy for advanced or metastatic disease.

4.2 Exclusion Criteria 1. Participants with EGFR, ALK, ROS-1, or known B-rapidly accelerated fibrosarcoma proto-oncogene (BRAF V600E) mutations that are sensitive to available targeted therapy. 2. Untreated CNS metastases. 3. Leptomeningeal metastases (carcinomatous meningitis). 4. Concurrent malignancy requiring treatment or history of prior malignancy active within 2 years prior to randomization (i.e., participants with a history of prior malignancy are eligible if treatment was completed at least 2 years before randomization and the participant has no evidence of disease). 5. Prior treatment with an anti-programmed cell death protein 1 (PD-1), anti-programmed death-ligand 1 (PD-L1), anti-programmed death-ligand 2 (PD-L2), or anti-cytotoxic T-lymphocyte-associated protein 4 (CTLA-4) antibody, or any other antibody or drug specifically targeting T-cell co-stimulation or checkpoint pathways.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Nivolumab + Relatlimab combined with two of the specified chemotherapies (carboplatin, cisplatin, paclitaxel, nab-paclitaxel, pemetrexed) administered on specified days of the treatment cycle. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:**

Cycles administered until disease progression, unacceptable toxicity, withdrawal of consent, or reaching the maximum specified duration for immunotherapy.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for chemotherapy-induced toxicities (e.g., anti-emetics), and systemic symptom management therapies. **Prohibited:** Other investigational anti-cancer therapies, targeted therapies for EGFR/ALK/ROS-1/BRAF, concurrent live vaccines, or other immunosuppressive medications (except for the management of immune-related adverse events).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Tumor biopsy/PD-L1 testing, Imaging, Baseline Labs, ECOG assessment.
Baseline	Day 0	Randomization, First dose of immunotherapy + PDCT.
Interim	Every 3-4 Weeks	IV Infusions, Safety Labs, PK/Biomarker blood collections, Assessment of irAEs.
Imaging	Every 6-8 Weeks	CT/MRI scans for tumor evaluation per RECIST 1.1.
End of Study	Follow-up	Final Vital Signs, Exit Interview, ongoing survival and progression follow-up.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. TRAEs Leading to Discontinuation Within 12 Weeks of First Dose in Part 1:** Percentage of participants with treatment related adverse events (TRAEs) leading to discontinuation within 12 weeks of first dose. Grading determined for severity according to the NCI CTCAE v5.0.
- 2. ORR Per RECIST v1.1 by BICR in Part 2:** Objective Response Rate (ORR) per RECIST v1.1 by BICR is defined as the number of participants in the randomized population who achieve a best overall response (BOR) of confirmed complete response (CR) or confirmed partial response (PR) based on BICR assessments divided by the number of participants in the population.

7.2 Secondary & Exploratory Endpoints

1. TRAEs Leading to Discontinuation in Part 1.
2. Number of Participants With a Treatment Related AEs in Part 1.
3. Number of Participants With Treatment Related SAEs in Part 1.
4. Progression-Free Survival (PFS) and Duration of Response (DOR).
5. PFS and ORR stratified by PD-L1 expression levels.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: AE is defined as any new untoward medical occurrence or worsening of a preexisting medical

condition in a clinical investigation participant administered study treatment, graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events (CTCAE) version 5.0. **Serious Adverse Events (SAEs):** Safety signals and severe TRAEs leading to discontinuation monitored closely; reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee appointed to review interim safety data and trial conduct.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population utilized for primary efficacy (ORR, PFS) comparisons. Safety population for all participants who received ≥ 1 dose of study drug. **Sample Size Justification:** $N=468$ participants, powered to detect a statistically significant improvement in ORR and assess safety when adding relatlimab to standard chemoimmunotherapy regimens. **Handling of Missing Data:** Standard censoring rules applied for time-to-event endpoints. Non-responders imputed for missing tumor response data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits by the sponsor to verify adherence to protocol, GCP, and patient safety. **Audit Trail:** Secure, validated Electronic Data Capture (EDC) system featuring eCRF locking mechanisms and electronic queries/data clarification forms.

11. Study Identifiers & Governance

Primary ID: CA224-104 **Registry Number:** NCT04623775 **Posting ID:** 902426380423784548 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** RELATIVITY-104 **Official Regulatory Title:** A Phase 2 Randomized Study of Relatlimab Plus Nivolumab in Combination With Chemotherapy vs. Nivolumab in Combination With Chemotherapy as First Line Treatment for Participants With Stage IV or Recurrent Non-small Cell Lung Cancer (NSCLC)

12. Clinical Framework

Primary Condition: Non-small Cell Lung Cancer, Recurrent Non-small Cell Lung Cancer, Metastatic Non-small Cell Lung Cancer (Preferred UMLS Name: Non-Small Cell Lung Carcinoma). **Diagnosis Threshold:** Histologically confirmed advanced NSCLC without targetable driver mutations (EGFR/ALK/ROS-1/BRAF) **Medical**

Methodology: First-line systemic chemoimmunotherapy (LAG-3/PD-1 inhibition + Platinum Doublet) **Optimization Target:** Improvement of anti-tumor immune response, ORR, and evaluation of safety profiles (TRAEs)

13. Study Procedures & Methodology

Management Pathway: First-line systemic oncology clinical pathway for advanced NSCLC. **Education Protocol (HCP):** Site initiation visits and standardized algorithms for the early identification and management of immune-related adverse events (irAEs). **Education Protocol (Patient):** Informed consent processes detailing treatment schedules, chemotherapy toxicities, and the importance of promptly reporting irAE symptoms. **Quality Audit Mechanism:** Continuous monitoring of protocol adherence, RECIST 1.1 evaluations audited by Blinded Independent Central Review (BICR).

14. Observation & Follow-Up Schedule

Total Duration: Up to ~5 years (inclusive of screening, treatment, and long-term survival follow-up). **Onsite Visit Cadence:** Every 3-4 weeks concurrent with treatment administration, with imaging every 6-8 weeks. **Remote Monitoring Frequency:** Routine data monitoring between treatment cycles. **Checklist Review Interval:** Toxicity and safety assessment at Day 1 of every treatment cycle.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				